

ScholarPath — Testing Evidence Report

Purpose: Factual testing evidence to support the later authoring of *Chapter 5: Testing and Evaluation*. **Nature:** Evidence report only. This document does **not** constitute Chapter 5 and does **not** invent pass/fail results. **Prepared by:** QA / testing documentation review of the repository. **Date of analysis:** 2026-07-03 **Branch analysed:** `claude/scholarpath-testing-report-57we9m` **Repository:** `ma7moudalysalem/scholarpath`

0. How to read this report (evidence provenance)

Every result below is labelled with one of two provenance tags:

Tag	Meaning
Executed	The test command was actually run in this analysis environment and the output is reproduced/summarised here.
Code inspection only	The tests exist in the repository and were read, but could not be executed in this environment (see the constraints in §1 and §3). No pass/fail is asserted for these.

Environment constraints that shaped what could be executed:

- .NET SDK is not installed** in the analysis environment, and the required SDK (`10.0.201` , per `server/global.json`) could **not** be installed because the outbound egress policy blocks the Microsoft download host (`builds.dotnet.microsoft.com` → HTTP 403). Therefore **no backend dotnet test command could be executed**.
- Docker is not running** in the analysis environment. The integration test project relies on `Testcontainers.MsSql` (a real SQL Server container per test class), so **integration tests could not be executed** even had the SDK been present.
- Frontend Vitest unit tests were executed successfully** (Node 22 + npm are available; `registry.npmjs.org` is reachable).
- Playwright E2E tests could not be executed:** the pre-installed browser build does not match the version this project's Playwright pins, and the matching browser download host (`cdn.playwright.dev`) is egress-blocked (HTTP 403). The E2E suite is also designed to run against a **deployed** environment with seeded login credentials (see §3), which are not present here.

1. Testing Environment

Aspect	Detail
Backend framework	ASP.NET Core / .NET 10 (<code>net10.0</code>), Clean-Architecture solution <code>server/ScholarPath.slnx</code> (API, Application, Domain, Infrastructure)
Backend patterns under test	MediatR command/query handlers, FluentValidation validators, EF Core, Hangfire background jobs, SignalR hubs
Required backend SDK	.NET SDK <code>10.0.201</code> (<code>server/global.json</code> , <code>rollForward: latestFeature</code>)
Frontend framework	React 19 + TypeScript + Vite 8 (<code>client/</code>), TanStack Query, React Router 7, i18next (EN/AR, RTL)
Backend unit test framework	<code>xUnit</code> + <code>FluentAssertions</code> + <code>NSubstitute</code> + <code>Bogus</code> (fake data) + <code>MockQueryable.NSubstitute</code> ; EF Core <code>InMemory</code> and <code>Sqlite</code> providers; <code>coverlet</code> for coverage
Backend integration test framework	<code>xUnit</code> + <code>FluentAssertions</code> + <code>Microsoft.AspNetCore.Mvc.Testing</code> (<code>WebApplicationFactory<Program></code>) + <code>Testcontainers.MsSql</code> + <code>Respawn</code> (DB reset)
Frontend unit test framework	<code>Vitest 4</code> + <code>@testing-library/react</code> + <code>@testing-library/user-event</code> + <code>@testing-library/jest-dom</code> ; DOM via <code>happy-dom</code>
Frontend E2E framework	<code>Playwright</code> (<code>@playwright/test</code>), projects: Chromium, Firefox, mobile (Pixel 7)
Test database (unit)	EF Core InMemory / SQLite in-memory (no external DB)
Test database (integration)	SQL Server 2022 via Testcontainers (<code>mcr.microsoft.com/mssql/server:2022-latest</code>) — requires Docker
Commands executed in this analysis	<code>npm install</code> (client)  ; <code>npm test</code> → <code>vitest run</code> (client)  ; <code>dotnet test</code>  (SDK unavailable); <code>npx playwright test smoke.spec.ts</code>  (browser unavailable)

2. Existing Test Assets

Backend test projects live under `server/tests/` ; frontend tests under `client/src/test/` .

Area	Test File / Project	Test Type	Purpose
Backend — all unit tests	<code>server/tests/ScholarPath.UnitTests/</code> (~111 test classes)	Unit (xUnit)	Handler logic, validators, domain rules, background jobs, security utilities
Backend — all integration tests	<code>server/tests/ScholarPath.IntegrationTests/</code>	Integration (xUnit + Testcontainers)	Full-API endpoint round-trips against a real SQL Server
Authentication / Onboarding	<code>Auth/LoginCommandHandlerTests</code> , <code>RegisterCommandHandlerTests</code> , <code>SelectRoleCommandHandlerTests</code> , <code>EmailVerificationTests</code> , <code>JwtRs256Tests</code> , <code>MemorySsoStateStoreTests</code> , <code>EmailChangeValidatorTests</code> , <code>RequestEmailChangeCommandHandlerTests</code>	Unit	Login/lockout, registration, role selection + onboarding gating, email verification, RS256 JWT, SSO state nonce
Profile & Account	<code>Profile/UpdateProfileCommandHandlerTests</code> , <code>UpdateProfileCommandValidatorTests</code> , <code>ChangePasswordCommandValidatorTests</code> , <code>ProfilePhotoUploadValidationTests</code>	Unit	Profile update logic + validation, password-change rules, photo upload validation
Scholarship Discovery	<code>Scholarships/ScholarshipBookmarksAndFeaturedTests</code> , <code>ScholarshipTests</code> , <code>Scholarships/ScholarshipAutoCloseJobTests</code>	Unit	Bookmarks list, featured list, localisation (AR), auto-close job; domain defaults
Local Scholarship Mgmt	<code>Scholarships/ScholarshipTests</code> , <code>AllowFreeScholarshipsSettingTests</code>	Unit	Scholarship domain state defaults, free-scholarship setting
External Scholarship	<code>Applications/ExternalIntentCommandHandlerTests</code>	Unit	External-listing "Intending" application, tracking-URL validation, re-application
Application Submission & Tracking	<code>Applications/CreateApplicationCommandHandlerTests</code> , integration <code>Applications/CreateApplicationCommandHandler</code>	Unit + Integration	Draft create/resume, status-transition state machine, active/terminal/read-only rules
Application Responding	<code>ScholarshipProviderReviewRequests/*</code> (Start, Accept, Cancel, ConfirmHoldAndComplete, RejectAndExpire, notifications), <code>ScholarshipProviderReviews/ApplicationStatusChangedEventHandlerTests</code>	Unit	Provider responding workflow, status-change events
Consultant Discovery & Booking	<code>ConsultantBookings/*</code> (BookingQueryHandler, ConsultantQueryHandler, ConsultantReadService, Reschedule, MeetingNoShow, BookingReminderJob, SessionRecording, RequestBookingMasterSwitch) + integration endpoint tests (Request/Accept/Reject/Cancel/MarkNoShow)	Unit + Integration	Booking lifecycle, availability, consultant discovery/read, no-show, reminders
Community	<code>Community/*</code> (CreatePost, UpdatePost, DeletePost, CreateReply, FlagPost, ToggleVote, ToggleBookmark, ForumReplyCreatedEventHandler) + <code>TagInput.test.tsx</code> (frontend)	Unit	Post/reply CRUD, role permissions, tags, votes, bookmarks, flagging
Messaging	<code>Chat/SendMessageCommandHandlerTests</code> , <code>Chat/GetConversationsQueryHandlerTests</code> , <code>Chat/ChatContactReadServiceTests</code> , integration <code>Chat/BlockEnforcementIntegrationTests</code> , <code>Hubs/PresenceTrackerTests</code>	Unit + Integration	Send/list conversations, contact search filters, block enforcement, presence
Resources Hub	<code>Resources/*</code> (CreateResource, SearchResources, ResourcePublishWorkflow, FeatureResource, ResourceStudentActions)	Unit	Resource publish workflow (draft → review → publish), search/filter, student actions
AI Advisory	<code>Ai/*</code> (AskChatbotRedaction, EligibilityVerdict, AiCostGate, LogRecommendationClick, KnowledgeBaseIndexer), integration <code>Ai/RedactionAuditSamplingJobTests</code>	Unit + Integration	PII redaction, eligibility verdict, cost gating, recommendation click logging, redaction audit sampling
Dashboards / Role-Based Views	Frontend <code>ProfileMenu.test.tsx</code> (role switcher); E2E route-guard specs <code>auth.spec.ts</code>	Unit (FE) + E2E	Role switching UI; route guards per role
Notifications	<code>Notifications/*</code> (NotificationCatalog, NotificationDispatcher, NotificationPreferences, ReminderJobs), integration <code>Notifications/EmailDeliveryIntegrationTests</code>	Unit + Integration	Catalog, dispatch, preference matrix, reminder jobs, email delivery + idempotency
Admin Oversight	<code>Admin/*</code> (ChangeUserRole, SetUserStatus, UserAdministration, ExportUsersCsv, ReviewOnboardingRoleGrant, ReviewUpgradeRequestValidator, SendBroadcast, LowRatedCompanies), integration <code>Admin/AtRiskUserChipTests</code> , <code>UpgradeRequests/SubmitConsultantUpgradeRequestCommandHandlerTests</code>	Unit + Integration	Role/status admin, onboarding review, broadcasts, at-risk/low-rated flags, upgrade requests
Security, Privacy & Audit	<code>Security/*</code> (FieldEncryption, FieldEncryptionConverter, FileScanning), <code>Audit/*</code> (AuditBehavior, DataExportJob, DataDeleteJob), <code>Common/ReviewerNameMaskTests</code> , <code>Documents/DocumentVaultTests</code>	Unit	Field encryption round-trip/tamper, file scanning, audit logging, GDPR export/delete, reviewer-name masking, document vault access
Frontend — unit/component	<code>client/src/test/*.test.tsx</code> (Home, ProfileMenu, ResetPassword, TagInput, usePlatformStatus)	Unit (Vitest)	Page render, role-switcher, password rules, tag input, platform-status hook
Frontend — E2E	<code>client/src/test/e2e/*.spec.ts</code> (14 specs)	E2E (Playwright)	Cross-module user journeys (see §3 and §6)

Area	Test File / Project	Test Type	Purpose
Removed-scope (financial/payment ent)	Payments/* , ProfitShare/* , FinancialConfig/* , Scholarships/{CreateScholarshipReviewFeeValidator , PaymentsEnabledMasterSwitch} , ConsultantBookings/{RefundCalculatorService , BookingPaymentSync} , ScholarshipProviderReviews/{Capture/Refund/RejectPayment/TimeoutRefundJob} , integration Payments/PaymentsIntegrationTests , e2e payments.spec.ts	Unit + Integration + E2E	Out of final scope — see \$Scope Alignment note below

Scope Alignment note

The repository still contains a substantial body of tests exercising **removed financial/payment scope** (Stripe payment intents, Stripe Connect, refunds, payouts, review fees, profit-share configuration/calculation, and the "payments-enabled" master switch). Per the final ScholarPath scope, these are **excluded** from the in-scope module evaluation in this report. They remain in the repository and are listed above and in §6 for completeness, but no in-scope pass/fail conclusion should be drawn from them.

Terminology used in this report (final scope): *Scholarship Provider* (the code and some routes still use "Company"), *Application Responding* (code namespace: `ScholarshipProviderReviewRequests`), *In Assessment* (vs. legacy "UnderReview"), *Local Scholarships*, *External Scholarships*, *Resources Hub*, *Admin-approved Consultants*. Where a test/file name uses legacy terminology, the original name is preserved for traceability.

3. Executed Test Commands

Command	Location	Result	Notes
npm install	client /	Executed — success (exit 0)	Dependencies installed from <code>registry.npmjs.org</code> .
npm test (vitest run)	client /	Executed — PASS: 5 files, 17/17 tests passed	Duration ~3.0 s. One non-fatal <code>ECONNREFUSED 127.0.0.1:3000</code> was logged (a stray/unmocked network attempt) but did not fail any test; the run exited 0.
npx vitest run --reporter=verbose	client /	Executed — 17/17 passed	Used to capture individual test names (see §4).
dotnet test ... ScholarPath.UnitTests	server /	NOT executed	<code>dotnet</code> not installed; SDK <code>10.0.201</code> download blocked by egress policy (<code>builds.dotnet.microsoft.com</code> → 403).
dotnet test ... ScholarPath.IntegrationTests	server /	NOT executed	Requires .NET SDK and Docker (Testcontainers SQL Server). Neither available here.
npx playwright test smoke.spec.ts --project=chromium	client /	NOT executed (browser launch failed)	Pre-installed browser build (v1194) ≠ version pinned by this Playwright (v1217); matching browser download blocked (<code>cdn.playwright.dev</code> → 403).
npx playwright install chromium-headless-shell	client /	Failed (egress 403)	<code>cdn.playwright.dev</code> host not permitted by egress policy.

Reference commands defined by the project (`docs/TESTING.md` , CI workflows) — provided for Chapter 5 reproducibility, not executed here:

- Backend, all: `cd server && dotnet test`
- Backend, unit only (CI): `dotnet test server/tests/ScholarPath.UnitTests/ScholarPath.UnitTests.csproj --configuration Release --collect:"XPlat Code Coverage"`
- Backend, integration (CI, push to `main` / `integration` only, `continue-on-error`): `dotnet test server/tests/ScholarPath.IntegrationTests/ScholarPath.IntegrationTests.csproj --configuration Release`
- Frontend, unit: `cd client && npm run test`
- Frontend, E2E (needs live target + seeded creds): `cd client && npm run test:e2e`

4. Unit Testing Evidence

4a. Frontend unit tests — Executed (Vitest), 17/17 passed

Module	Tested Functionality	Test File(s)	Result	Notes
Public site / Discovery entry	Home page renders an <code><h1></code> hero heading	<code>Home.test.tsx</code> (1)	PASS (Executed)	Renders under Router + React Query.
Dashboards / Role-based views	Role switcher: hidden for single-role, shown for dual-role, calls switch-role API	<code>ProfileMenu.test.tsx</code> (3)	PASS (Executed)	Mocks notifications + auth APIs.
Authentication (reset password)	Invalid-link state; rejects short / missing-uppercase-digit-special / mismatched; accepts valid	<code>ResetPassword.test.tsx</code> (5)	PASS (Executed)	Password rules mirror Register.
Community	Tag commit + slugify on Enter; case-insensitive dedupe; caps at 5 tags; chip removal	<code>TagInput.test.tsx</code> (4)	PASS (Executed)	Cap constant <code>MAX_TAGS_PER_POST = 5</code> .
Platform status (payments toggle surface)	Safe defaults before resolve; reflects <code>paymentsEnabled</code> true/false; fallback true on error	<code>usePlatformStatus.test.tsx</code> (4)	PASS (Executed)	Confirms the client honours a server "payments disabled" flag.

4b. Backend unit tests — Code inspection only (not executed)

Counted by `[Fact]` / `[Theory]` attributes: **≈696 unit test methods across ~111 test classes**. Actual executed case count would be **higher**, because `[Theory]` methods expand into multiple cases via `[InlineData]` . **No pass/fail is asserted** — the suite was not run.

Module	Tested Functionality (representative)	Test File(s)	Result	Notes
Authentication / Onboarding	Valid login → tokens; wrong password/unknown email → conflict; logout on 5th failure; role selection activates student / queues provider onboarding / persists consultant profile; onboarding blocked without documents; RS256 JWT; SSO state nonce	Auth/LoginCommandHandlerTests (5), SelectRoleCommandHandlerTests (20), RegisterCommandHandlerTests (3), EmailVerificationTests (7), JwtRs256Tests (4), MemorySsoStateStoreTests (3), EmailChangeValidatorTests (5), RequestEmailChangeCommandHandlerTests (3)	Code inspect ion only	Strong branch coverage on auth flows.
Profile & Account	Profile update persistence; extensive field validation (25 validator cases); password-change rules; photo upload validation	Profile/UpdateProfileCommandHandlerTests (7), UpdateProfileCommandValidatorTests (25), ChangePasswordCommandValidatorTests (9), ProfilePhotoUploadValidationTests (6)	Code inspect ion only	—
Scholarship Discovery	Bookmarks (own-only, ordering, orphan drop, AR localisation, forbidden-when-anon); featured (open+non-deleted only, ordering, limit, AR); auto-close job	Scholarships/ScholarshipBookmarksAndFeaturedTests (11), ScholarshipAutoCloseJobTests (5), ScholarshipTests (4)	Code inspect ion only	No dedicated scholarship search/filter/pagination query-handler test found (see §6).
Local Scholarship Mgmt	Scholarship domain defaults (Draft), free-scholarship setting behaviour	ScholarshipTests (4), AllowFreeScholarshipsSettingTests (4)	Code inspect ion only	Local scholarship create/edit lifecycle mostly covered via integration + provider-review suites.
External Scholarship	External-listing intending application; not-found; rejects in-app listing; duplicate-active conflict; re-apply after terminal; tracking-URL validation	Applications/ExternalIntentCommandHandlerTests (12)	Code inspect ion only	—
Application Submission & Tracking	Draft create/resume/no-duplicate; fresh draft after terminal; status-transition state machine (allowed/invalid transitions, active/terminal/read-only); ownership → NotFound/Conflict	Applications/CreateApplicationCommandHandlerTests (19)	Code inspect ion only	Rich state-machine coverage.
Application Responding	Provider responding request: create + (fee path) capture intent; free path skips payment → Pending; idempotent double-click; rejects own-scholarship / closed / no-fee; start/accept/cancel/confirm/reject-expire; status-changed event	ScholarshipProviderReviewRequests/* (Start 8, Accept 5, Cancel 6, ConfirmHoldAndComplete 5, RejectAndExpire 5, notif factory 2), ScholarshipProviderReviews/ApplicationStatusChangedEventHandlerTests (3)	Code inspect ion only	Fee/Stripe branches are removed-scope; the free/no-payment responding path is in-scope.
Consultant Discovery & Booking (no payment)	My/consultant bookings (own-only, soft-delete excluded, forbidden); booking-by-id authorisation matrix; availability round-trip; reschedule; no-show; reminder jobs; consultant read/discovery projections	ConsultantBookings/BookingQueryHandlerTests (16), ConsultantReadServiceTests (19), ConsultantQueryHandlerTests (5), RescheduleBookingCommandHandlerTests (6)+validator (3), MeetingNoShowTests (8), BookingReminderJobTests (9), SessionRecordingTests (7), RequestBookingMasterSwitchTests (2)	Code inspect ion only	Payment-coupled files (RefundCalculatorService , BookingPaymentSync) are removed-scope.
Community	Student creates post; consultant/provider cannot; tag normalisation + >5 tags error; reply/delete/update; flag; vote toggle; bookmark toggle; forum-reply event	Community/CreatePostCommandHandlerTests (5), UpdatePost (5), DeletePost (4), CreateReply (3), FlagPost (3), ToggleVote (3), ToggleBookmark (3), ForumReplyCreatedEventHandler (3)	Code inspect ion only	Role-permission matrix covered.
Messaging	Blocked-user send → conflict; conversation listing; contact search filters (excludes self/inactive/soft-deleted/blocked, name match, projection, limit, ordering); presence tracking	Chat/SendMessageCommandHandlerTests (1), GetConversationsQueryHandlerTests (5), ChatContactReadServiceTests (11), Hubs/PresenceTrackerTests (7)	Code inspect ion only	Only 1 unit test on the send-message handler ; real-time SignalR broadcast not unit-tested (see §6).
Resources Hub	Publish workflow (draft → pending → publish; admin direct publish; approve/reject; author-bio gate); search (published-only, category, title match); create; feature; student actions	Resources/ResourcePublishWorkflowTests (7), SearchResourcesQueryHandlerTests (3), CreateResourceCommandHandlerTests (4), FeatureResourceCommandHandlerTests (4), ResourceStudentActionsTests (4)	Code inspect ion only	—
AI Advisory	PII redaction patterns; empty/long-message guards; eligibility verdict; AI cost gate; recommendation-click logging; knowledge-base indexer	Ai/AskChatbotRedactionTests (6), EligibilityVerdictTests (7), AiCostGateTests (5), LogRecommendationClickTests (9), KnowledgeBaseIndexerTests (1)	Code inspect ion only	KB indexer has only 1 case; RAG retrieval quality not unit-tested (expected).
Notifications	Preference matrix (defaults, disable/enable/re-enable, no-leak, auth); catalog; dispatcher; reminder jobs	Notifications/NotificationPreferencesTests (11), NotificationCatalogTests (8), ReminderJobsTests (9), NotificationDispatcherTests (3)	Code inspect	—

Module	Tested Functionality (representative)	Test File(s)	Result	Notes
			ion only	
Admin Oversight	Change role; set status; user administration; CSV export; onboarding role-grant review; upgrade-request validator; broadcasts; low-rated provider flag	Admin/ChangeUserRoleCommandTests (5), SetUserStatusCommandTests (6), UserAdministrationTests (5), ExportUsersCsvQueryHandlerTests (4), ReviewOnboardingRoleGrantTests (4), ReviewUpgradeRequestValidatorTests (4), SendBroadcastCommandHandlerTests (4), LowRatedCompanies/LowRatedCompaniesTests (8), UpgradeRequests/SubmitConsultantUpgradeRequestCommandHandlerTests (12)	Code inspection only	—
Security, Privacy & Audit	Field encryption round-trip/null/nonce-randomness/tamper/short-envelope/wrong-key; key-provider guards; EF value converter; file scanning; audit behaviour pipeline; GDPR export (all tables, audit entry, idempotent); GDPR delete; reviewer-name masking; document vault access	Security/FieldEncryptionTests (14), FieldEncryptionConverterTests (3), FileScanningTests (5), Audit/AuditBehaviorTests (4), DataExportJobTests (5), DataDeleteJobTests (7), Common/ReviewerNameMaskTests (5), Documents/DocumentVaultTests (15)	Code inspection only	Strong privacy/security coverage.
Domain smoke	Entity defaults (user Unassigned, scholarship Draft, tracker Active/status)	Smoke/SmokeTests (4)	Code inspection only	—
Streaming	Domain-event publishing handler	Streaming/DomainEventPublishingHandlerTests (5)	Code inspection only	—
Common	Country normaliser	Common/CountryNormalizerTests (5)	Code inspection only	—

5. Integration Testing Evidence

Code inspection only (not executed). Counted by `[Fact]` / `[Theory]` : **≈35 integration test methods**. These use `WebApplicationFactory<Program>` + `Testcontainers.MsSql` (real SQL Server) + `Respawn` , and cannot run without Docker. In CI they run **only on push to main / integration** and with `continue-on-error: true` (informational, non-blocking).

Scenario	Components Integrated	Test File(s)	Result	Notes
Start application via API	HTTP endpoint → MediatR → EF Core → SQL Server	<code>Applications/CreateApplicationCommandHandler</code> (5): valid → 201; duplicate-active → 409; closed → 409; external → 409; empty id → 400	Code inspection only	In-scope.
Messaging block enforcement	Chat endpoint → block rules → DB	<code>Chat/BlockEnforcementIntegrationTests</code> (4): no-block → 200; recipient-blocked-sender → 409; sender-blocked-recipient → 409; after-unblock → 200	Code inspection only	In-scope.
Consultant booking endpoints	API → booking domain → DB	<code>ConsultantBookings/RequestBookingEndpointTests</code> (1), <code>AcceptBookingEndpointTests</code> (1), <code>RejectBookingEndpointTests</code> (1), <code>CancelBookingEndpointTests</code> (1), <code>MarkNoShowEndpointTests</code> (1)	Code inspection only	In-scope (booking lifecycle).
Admin at-risk user chip	Admin search → risk-flag join → DTO	<code>Admin/AtRiskUserChipTests</code> (3): true when flag row; false when none; false when flag=false	Code inspection only	In-scope.
Email delivery	Notification dispatch → email channel → persistence	<code>Notifications/EmailDeliveryIntegrationTests</code> (3): correct recipient; idempotency dedupe; persists dispatch-succeeded row	Code inspection only	In-scope.
AI redaction audit sampling	Hangfire job → interactions → sampling	<code>Ai/RedactionAuditSamplingJobTests</code> (3): idempotent for fully-sampled month; only samples previous month	Code inspection only	In-scope.
Scholarship API	<code>WebApplicationFactory</code> + Testcontainers factory	<code>Scholarships/ScholarshipApiTests</code>	No test methods	⚠ File defines only the <code>ScholarshipApiFactory</code> (32 lines) — 0 executable tests.
Scholarship applications (extended)	<code>WebApplicationFactory</code> + Testcontainers + seed	<code>Applications/ScholarshipApplicationsIntegrationTests</code>	No test methods	⚠ File defines only the factory + student seed helper (117 lines) — 0 executable tests.
Payments (removed scope)	Stripe/payment endpoints → DB	<code>Payments/PaymentsIntegrationTests</code> (12)	Code inspection only	Removed scope — excluded from evaluation.

6. Failed / Skipped / Missing Tests

No test *failures* can be reported for the backend or E2E suites because **they were not executed** (see §1/§3). The executed frontend Vitest suite reported **0 failures**. The items below are structural gaps and execution blockers found by inspection.

Area	Issue	Impact	Recommended Action
Backend execution	.NET SDK <code>10.0.201</code> unavailable and un-installable here (egress-blocked); <code>dotnet test</code> not run	No executed backend pass/fail evidence in this environment	Run <code>dotnet test</code> on a machine/CI with .NET 10 SDK; attach <code>.trx</code> + coverage <code>.cobertura.xml</code> as Chapter 5 evidence
Integration execution	Requires Docker (Testcontainers SQL Server); Docker not running here	No executed integration evidence; CI also runs them <code>continue-on-error</code> (non-blocking)	Run integration job with Docker; capture results; consider promoting critical in-scope endpoints to a blocking check
E2E execution	Playwright browser version mismatch (v1194 present vs v1217 required) + download host blocked; suite targets a deployed URL with seeded creds	E2E cannot run locally here; many specs self-skip without credentials	Run E2E via the <code>e2e.yml</code> workflow against a deployed environment with <code>E2E_*</code> secrets; archive the Playwright HTML/JUnit report
Integration: Scholarship API	<code>Scholarships/ScholarshipApiTests.cs</code> contains only a factory, 0 tests	Public scholarship endpoints have no integration coverage despite an existing harness	Add integration tests (list/detail/create/close) using the existing <code>ScholarshipApiFactory</code>
Integration: Scholarship applications	<code>Applications/ScholarshipApplicationsIntegrationTests.cs</code> contains only a factory + seed, 0 tests	Intended extended application integration tests are not implemented	Implement the intended endpoint tests on top of the seeded factory
Scholarship Discovery (search)	No dedicated search/filter/pagination query-handler unit test (only bookmarks/featured/auto-close)	The primary discovery experience (keyword/eligibility/country/level filters) is weakly tested	Add unit tests for the scholarship search/filter/sort/pagination handler
Messaging (real-time)	SignalR hub message broadcast/delivery is not integration-tested; send-message handler has 1 unit test	Real-time chat delivery, typing/read state untested end-to-end	Add a SignalR <code>HubConnection</code> integration test for send – receive; add handler unit tests (self-send, non-contact, length limits)
Dashboards / Role-based views	No backend dashboard query-handler tests ; dashboards covered only by (non-executable-here) E2E route guards	Dashboard aggregation/role-scoping logic unverified at unit level	Add unit tests for each role's dashboard aggregation query
E2E self-skips	~23 <code>test.skip</code> / <code>test.fixme</code> guards across specs (e.g., admin 2, analytics 3, applications 2, booking 2, community 2, notifications 2, profile 2, scholarships 2); auth specs also self-skip via <code>hasCreds(role)</code> when <code>E2E_*</code> secrets are absent	Reported E2E pass counts can be misleadingly low if run without seeded credentials	In Chapter 5, always report skipped counts alongside passed; provision credentials for the target environment
Frontend unit breadth	Only 5 frontend unit test files (17 cases) for a large React app	Most pages/components rely on E2E only	Add component tests for key forms (Login/Register/Onboarding), application submission, booking, messaging UI
Coverage gate	<code>docs/TESTING.md</code> states a ≥70% target "enforced by CI", but <code>ci.yml</code> collects coverage without a failing threshold gate (the <code>/p:Threshold=70</code> gate is marked roadmap)	Coverage claim is aspirational, not enforced	Either enforce the threshold in CI or state the measured coverage % (from a real run) in Chapter 5
Removed-scope tests present	Payment/Stripe/refund/profit-share/fee tests still in the repo (unit + integration + <code>payments.spec.ts</code>)	Risk of accidentally citing out-of-scope results as product evidence	Exclude from Chapter 5 evaluation; optionally remove or quarantine to match final scope

7. Recommended Chapter 5 Test Cases

These are **recommendations to document/execute later** — not executed results. They target in-scope modules, prioritising the gaps in §6. "Related Requirement" cites identifiers actually seen in the repo where available (e.g., `FR-AUTH-13` , E2E tag `PB-001 / T-018`); otherwise a descriptive use-case name is used.

Test Case ID	Module	Actor	Scenario	Preconditions	Test Steps	Expected Result	Related Req / Use Case
TC-01	Authentication	Registered student	Successful login issues tokens	Active verified account exists	Submit valid email+password to <code>/api/auth/login</code>	200 + access/refresh tokens; user in Active state	Login use case (cf. <code>LoginCommandHandlerTests</code>)
TC-02	Authentication	Registered student	Account locks after repeated failures	Account exists	Submit wrong password 5×	5th attempt locks account; subsequent login rejected	Lockout (cf. <code>Handle_FifthFailedAttempt_LocksAccount</code>)
TC-03	Onboarding	New Scholarship Provider	Provider selection enters onboarding queue	Authenticated, role Unassigned, verification docs uploaded	Select "Scholarship Provider" role with required fields + docs	Account queued for admin approval; active admins notified with deep link	<code>SelectRoleCommandHandlerTests</code> ; onboarding UC
TC-04	Onboarding	New Provider (no docs)	Provider onboarding blocked without documents	Authenticated, no verification docs	Attempt provider role selection	Rejected — verification documents required	<code>ScholarshipProvider_submission_blocked_when_no_verification_documents</code>
TC-05	Profile	Student	Update profile persists and validates	Authenticated	Submit valid profile update; then submit invalid fields	Valid update persists; invalid fields return validation errors	<code>UpdateProfileCommand*Tests</code>
TC-06	Scholarship Discovery	Student	Search/filter local scholarships	≥1 Open scholarship exists	Search by keyword + country + level with pagination	Only matching, Open, non-deleted results; correct order + page size	Gap — no current handler test
TC-07	Scholarship Discovery	Student	Bookmark then list bookmarks	Authenticated; scholarship exists	Bookmark a scholarship; open bookmarks	Bookmark appears newest-first; localised (AR) when language=AR	<code>ScholarshipBookmarksAndFeaturedTests</code>
TC-08	Local Scholarship Mgmt	Scholarship Provider	Create → publish → auto-close scholarship	Provider approved	Create scholarship, publish, advance past deadline	New scholarship defaults Draft; publishes Open; auto-close job closes after deadline	<code>ScholarshipTests</code> , <code>ScholarshipAutoCloseJobTests</code>
TC-09	External Scholarship	Student	Track an external scholarship	External listing exists	Create intending application with valid tracking URL	Intending application created; invalid tracking URL rejected; duplicate active → conflict	<code>ExternalIntentCommandHandlerTests</code>
TC-10	Application Submission	Student	Submit and resume application drafts	Open scholarship; authenticated	Start application; re-open; submit	Draft created once (resumed, not duplicated); valid status transitions enforced; terminal → read-only	<code>CreateApplicationCommandHandlerTests</code> (unit + integration)
TC-11	Application Responding	Scholarship Provider	Respond to a submitted application (free path)	Submitted application on provider's scholarship; payments disabled	Start responding; move to In Assessment; complete	Free path skips payment and lands in Pending/In Assessment; own-scholarship & closed rejected; idempotent on double-click	<code>StartScholarshipProviderReviewRequestCommandHandlerTests</code>
TC-12	Consultant Booking (no payment)	Student ↔ Consultant	Request → accept → no-show lifecycle	Admin-approved consultant with availability	Student requests slot; consultant accepts; mark no-show	Booking created (201) and confirmed (204); authorisation matrix enforced; no-show recorded	Booking endpoint integration tests
TC-13	Community	Student / Consultant	Post creation permissions + tags	Authenticated users of each role	Student creates post with 6 tags; consultant/provider attempt to post	Student post succeeds with tags normalised and capped at 5; consultant/provider rejected	<code>CreatePostCommandHandlerTests</code> , <code>TagInput.test.tsx</code>
TC-14	Messaging	Two students	Send message with block enforcement + real-time delivery	Both active; contact relationship	A sends to B (200); B blocks A; A resends (409); real-time receipt on B	Correct 200/409 per block state; message delivered over SignalR to online recipient	<code>BlockEnforcementIntegrationTests</code> ; real-time gap
TC-15	Resources Hub	Author / Admin	Resource publish workflow	Author has bio; admin available	Submit complete draft (→ pending); admin approves (→ published); reject returns to draft	Workflow transitions enforced; incomplete draft / missing bio blocked; admin submit publishes directly	<code>ResourcePublishWorkflowTests</code>
TC-16	AI Advisory	Student	Chatbot redacts PII before processing	AI enabled; within cost budget	Send message containing PII; send over-budget request	PII redacted to tokens; empty/over-long rejected; cost gate blocks when budget exceeded	<code>AskChatbotRedactionTests</code> , <code>AiCostGateTests</code>
TC-17	Dashboards / Role-based	Each role	Route guards + role-scoped dashboard	Unauthenticated + each authenticated role	Hit role routes unauthenticated; then as each role	Unauthenticated redirected to <code>/login</code> ; each role sees only its dashboard data	E2E <code>auth.spec.ts</code> (PB-001 / T-018); dashboard-handler gap
TC-18	Notifications	Student	Preference matrix controls delivery	Authenticated	Disable a channel; trigger a notification; re-enable	Disabled channel suppressed; others delivered; email dispatch idempotent	<code>NotificationPreferencesTests</code> , <code>EmailDeliveryIntegrationTests</code>
TC-19	Admin Oversight	Admin	Approve onboarding + manage users	Admin; pending onboarding request	Approve provider onboarding; change a user role; set status; flag low-rated provider	Onboarding approved; role/status changes persisted + audited; low-rated flag surfaced; at-risk chip reflects flag	<code>Admin/*</code> , <code>AtRiskUserChipTests</code>

Test Case ID	Module	Actor	Scenario	Preconditions	Test Steps	Expected Result	Related Req / Use Case
TC-20	Security / Privacy / Audit	Student / System	Field encryption + GDPR export/delete	Sensitive fields stored; authenticated user	Round-trip encrypt/decrypt; tamper ciphertext; request data export then deletion	Round-trip returns plaintext; tamper rejected; export covers all related tables + writes audit entry (idempotent); delete removes personal data	<code>FieldEncryptionTests</code> , <code>DataExportJobTests</code> , <code>DataDeleteJobTests</code>

8. Testing Results Summary

Executed rows carry real pass/fail. **Code-inspection-only** rows report the *number of defined tests* discovered; their Passed/Failed/Skipped are shown as `—` because the suites were **not run** in this environment. `[Theory]` methods expand to more cases at runtime, so backend "Total" is a **lower bound** on executable cases.

Test Category	Total Tests	Passed	Failed	Skipped	Notes
Frontend unit (Vitest)	17	17	0	0	Executed — 5 files, exit 0, ~3.0 s
Backend unit (xUnit)	≈696 methods (~111 classes)	—	—	—	Code inspection only — not executed (no .NET 10 SDK; download egress-blocked). Includes removed-scope payment/profit-share/financial tests that should be excluded from evaluation
Backend integration (xUnit + Testcontainers)	≈35 methods	—	—	—	Code inspection only — not executed (needs Docker + SDK). 2 integration files contain factories with 0 tests
Frontend E2E (Playwright)	≈61 <code>test()</code> declarations across 14 specs (more at runtime via parametrised loops)	—	—	~23 <code>skip</code> / <code>fixme</code> + credential-gated self-skips	Not executed — browser version mismatch + egress-blocked download; suite targets a deployed URL with seeded creds
In-scope executed evidence	17	17	0	0	Only the frontend Vitest suite was actually executed here

Removed-scope tests present in the repository (excluded from the evaluation above)

Suite	Approx. count	Location
Payments (Stripe intents, Connect, refunds, payouts, webhooks)	~45 unit methods	<code>UnitTests/Payments/</code>
Profit share (calculator, resolver, config)	~18 unit methods	<code>UnitTests/ProfitShare/</code>
Financial config (calculator, rule resolver)	~16 unit methods	<code>UnitTests/FinancialConfig/</code>
Payment-coupled (booking refund/sync, provider-review payment/refund, review-fee validators)	additional methods embedded in booking, provider-review, and scholarship suites	various
Payments integration	~12 methods	<code>IntegrationTests/Payments/PaymentsIntegrationTests.cs</code>
Payments E2E	3 (2 skipped)	<code>client/src/test/e2e/payments.spec.ts</code>

Appendix A — Exact commands & environment facts used for this report

- `server/global.json` → SDK `10.0.201`, `rollForward: latestFeature`.
- `dotnet` not on PATH; install blocked: `builds.dotnet.microsoft.com` → HTTP 403 (egress policy).
- `docker` present but daemon not running → Testcontainers unavailable.
- Node `v22.22.2`, npm `10.9.7`; `registry.npmjs.org` reachable.
- `client/` — `npm install` OK; `npm test` → **Test Files 5 passed (5) / Tests 17 passed (17)**.
- Playwright: pre-installed browser `chromium_headless_shell-1194`; project requires `-1217`; `cdn.playwright.dev` → HTTP 403.
- No committed test-result artifacts (`.trx` , `coverage.cobertura.xml` , `junit.xml` , `lcov.info`) found in the repository.

End of testing evidence report. This document is evidence only and is not Chapter 5.